Medical Diagnosis Center

Project Description:

The Medical Diagnosis Center is a ServiceNow-based digital diagnostic test management system designed to streamline patient registration, diagnostic test selection, appointment booking, approval management, laboratory processing, report generation, notifications, security and administrative reporting. The solution uses ServiceNow custom tables, Service Portal pages and widgets, Business Rules, Flow Designer workflows, ACLs, notifications, reports and dashboards to provide a centralized diagnostic-center workflow.

Patients can browse the Test Catalog, provide patient information, select an appointment date and time, track appointment status and access authorized laboratory reports. Staff roles such as receptionists, laboratory technicians, doctors, lab managers and administrators support the operational lifecycle through configured ServiceNow functionality.

Scenario 1: Patient Registration and Diagnostic Test Selection

A patient opens the Medical Diagnosis Center Service Portal, enters required patient information and browses configured diagnostic tests. Test records can contain the test name, code, category, description, price, duration and active status. The patient then selects an appropriate test before booking.

Scenario 2: Appointment Booking, Approval and Processing

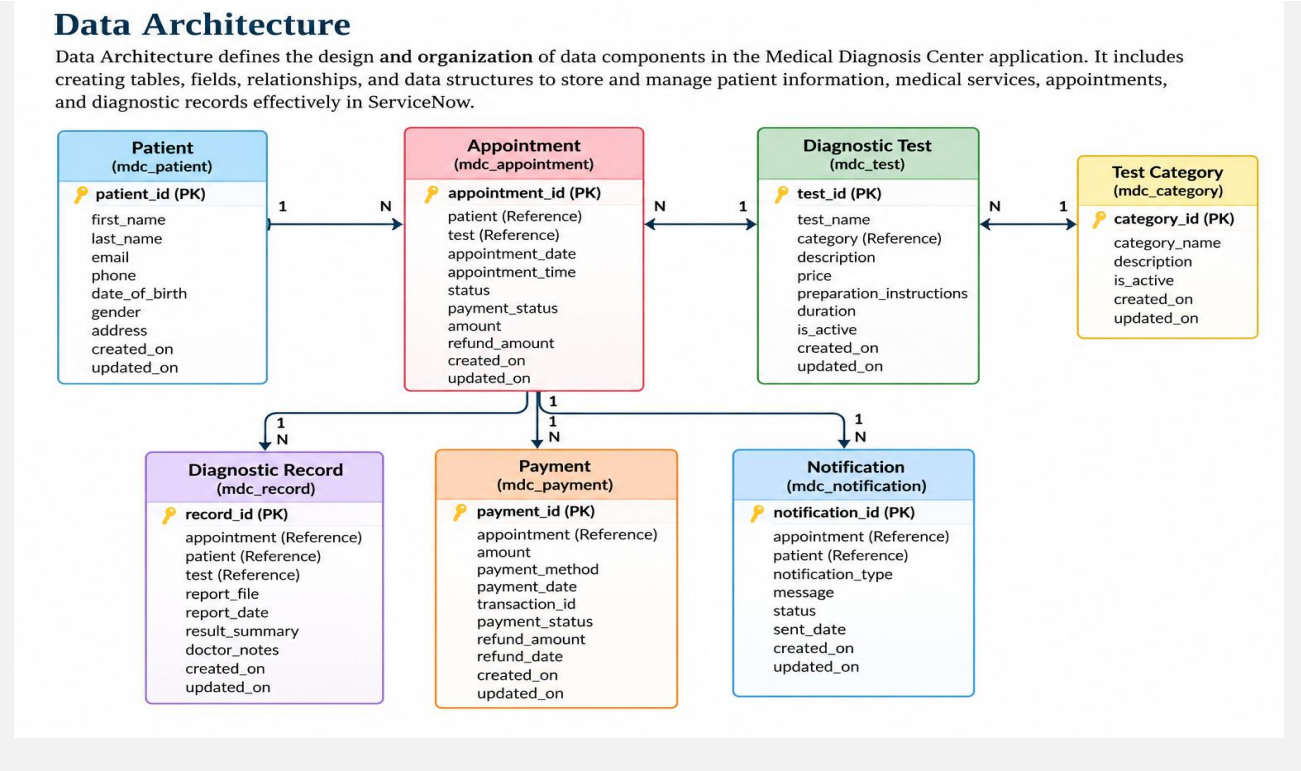
After selecting a test, the patient chooses an appointment date and time slot. The system is designed to validate mandatory fields, availability, duplicate and overlapping bookings, configured capacity and other booking rules. The appointment can move through Requested, Pending Approval, Confirmed, Sample Collected, Processing and Completed states.

Scenario 3: Report Availability and Secure Access

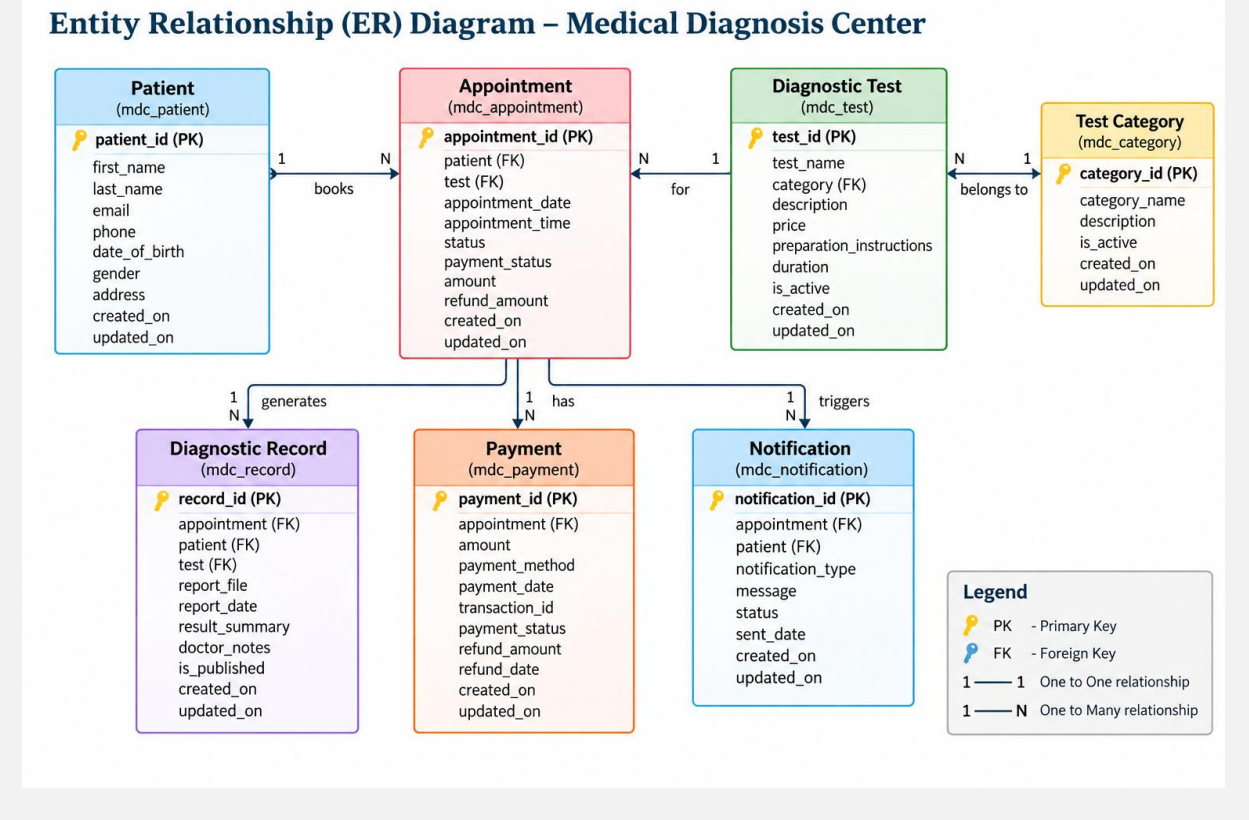
When diagnostic processing is completed, a report can be generated and linked to the appointment and patient. A configured notification can inform the patient

when the report becomes available. ACLs and patient-specific access controls are used to restrict report and application data access.

SERVICENOW ARCHITECTURE



Entity Relationship (ER) Diagram:



Pre-requisites:

1. ServiceNow instance
2. Service Portal enabled
3. Access to Studio / Scoped Application Development
4. Appropriate ServiceNow roles
5. Access to Email Notifications, Business Rules, ACLs and Reports & Dashboards
6. Access to Update Sets and application deployment configuration

Project WorkFlow:

7. 1. ServiceNow instance and application setup
8. 2. Create custom tables and relationships
9. 3. Design Service Portal pages and widgets
10. 4. Create diagnostic test catalog
11. 5. Implement patient registration
12. 6. Implement appointment booking and validation
13. 7. Configure approval and status workflow
14. 8. Configure laboratory processing and report generation
15. 9. Configure email notifications
16. 10. Configure payment/refund logic as enabled
17. 11. Configure ACL security and patient-specific access
18. 12. Create reports and Health Reports Dashboard
19. 13. Test the end-to-end workflow and validate Update Set deployment

Medical Diagnosis Center

Project Phases – ServiceNow

The development of the Medical Diagnosis Center is divided into five phases. Each phase covers a part of the work carried out during the project, from planning the application to testing and preparing the final demonstration.

Phase 1: Requirement Analysis & Planning

• The project begins with understanding the requirements of a diagnostic centre and identifying the problems that the application should address.

• The main requirements include patient registration, diagnostic test selection, appointment booking, appointment approval, laboratory processing and report access.

• The users involved in the process are patients, receptionists, laboratory technicians, doctors, lab managers and the system administrator.

• The application structure, database tables, relationships and overall workflow are planned before development.

• The ServiceNow instance, Service Portal and required development permissions are checked before starting the configuration.

Work covered in this phase: project description, use-case scenarios, system architecture, ER diagram and prerequisites.

Phase 2: Backend Development & Configurations

• The backend is developed by creating the tables needed to store patient, diagnostic test, appointment, report and payment details.

• Fields and reference relationships are configured so that appointment records are connected to the correct patient and diagnostic test.

• Business Rules and GlideRecord are used for server-side operations, including appointment checks, status updates and report-related actions.

• Appointment validation is configured to check required information and prevent duplicate or overlapping bookings based on the rules set for the application.

• Approval handling, Flow Designer actions and email notifications are configured to support the appointment process.

• Payment and refund processing are included where the payment feature is enabled.

Work covered in this phase: custom tables, field configuration, appointment validation, Business Rules, approval handling, Flow Designer and notifications.

Phase 3: UI/UX Development & Customization

• The Service Portal is set up to provide patients with a simple interface for using the diagnostic centre application.

• The Home Page and Test Catalog Page are arranged to help patients view the available diagnostic tests and their details.

• The Book Appointment Page allows patients to select a test, enter their details and choose an appointment date and time.

• The My Appointments Page displays booking information and the current appointment status.

• The Lab Reports Page provides access to reports that are available to the logged-in patient.

• The required Service Portal widgets are configured and the pages are styled to keep navigation clear and consistent.

Work covered in this phase: Service Portal pages, Medical Test Catalog, Appointment Booking, My Appointments and Lab Reports widgets.

Project Phases (Continued)

Security, testing and final project activities

Phase 4: Data Management, Security & Testing

• The records created in the application are checked to make sure that patient, test, appointment and report information is stored correctly.

• The relationships between the records are verified, particularly the links between patients, appointments, diagnostic tests and reports.

• Access Control Lists are configured to control access to tables and fields according to the user's role.

• Patient-specific access is checked so that users can view only the records permitted for them.

• The main workflows are tested, including registration, test selection, appointment booking, approval, laboratory processing, report generation and email notifications.

• Cancellation, refund and other optional features are tested when they are enabled. Test results and any issues found are recorded.

Work covered in this phase: data checks, ACL configuration, role-based access, functional testing and end-to-end testing.

Phase 5: Deployment, Documentation & Final Presentation

• After configuration and testing, the required application changes are captured in a ServiceNow Update Set.

• The Update Set is reviewed and committed before the configuration is moved to the target instance, where applicable.

• The application is checked in the target instance and basic smoke testing is carried out to confirm that the main functions are working.

• The final documentation is completed with the project explanation, table details, workflows, configuration steps and screenshots from the ServiceNow instance.

• A project demonstration is prepared to show the complete process, from patient registration and appointment booking to report availability.

• The project is then presented with its implemented features, test results and possible future improvements.

Work covered in this phase: Update Set preparation, deployment checks, documentation, screenshots and project demonstration.

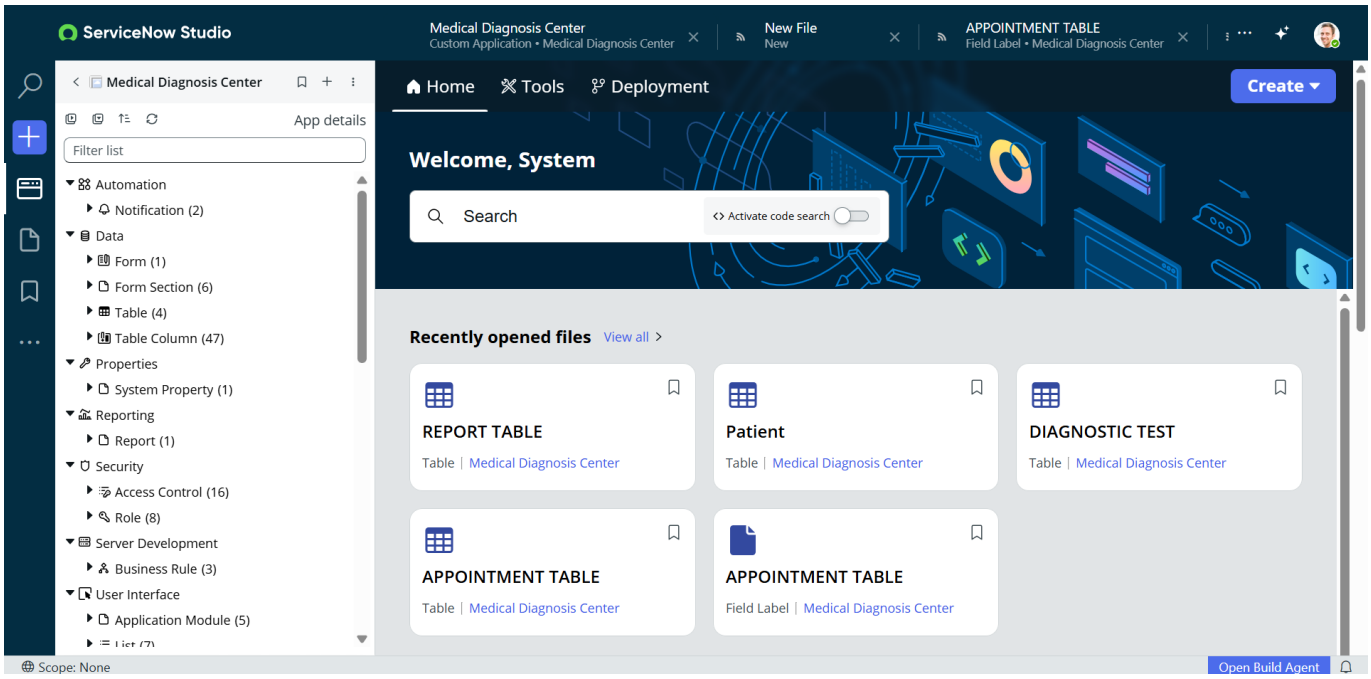
Project Conclusion

The Medical Diagnosis Center provides a single ServiceNow-based application for managing patient details, diagnostic tests, appointments and laboratory reports. The portal brings these activities into one workflow and uses configured automation, notifications and access controls to support the work of patients and diagnostic-centre staff.

The final report should reflect the features actually configured and tested in the ServiceNow instance. Any optional feature or test that has not yet been completed should be identified clearly.

Milestone 1: ServiceNow Instance and Application Setup

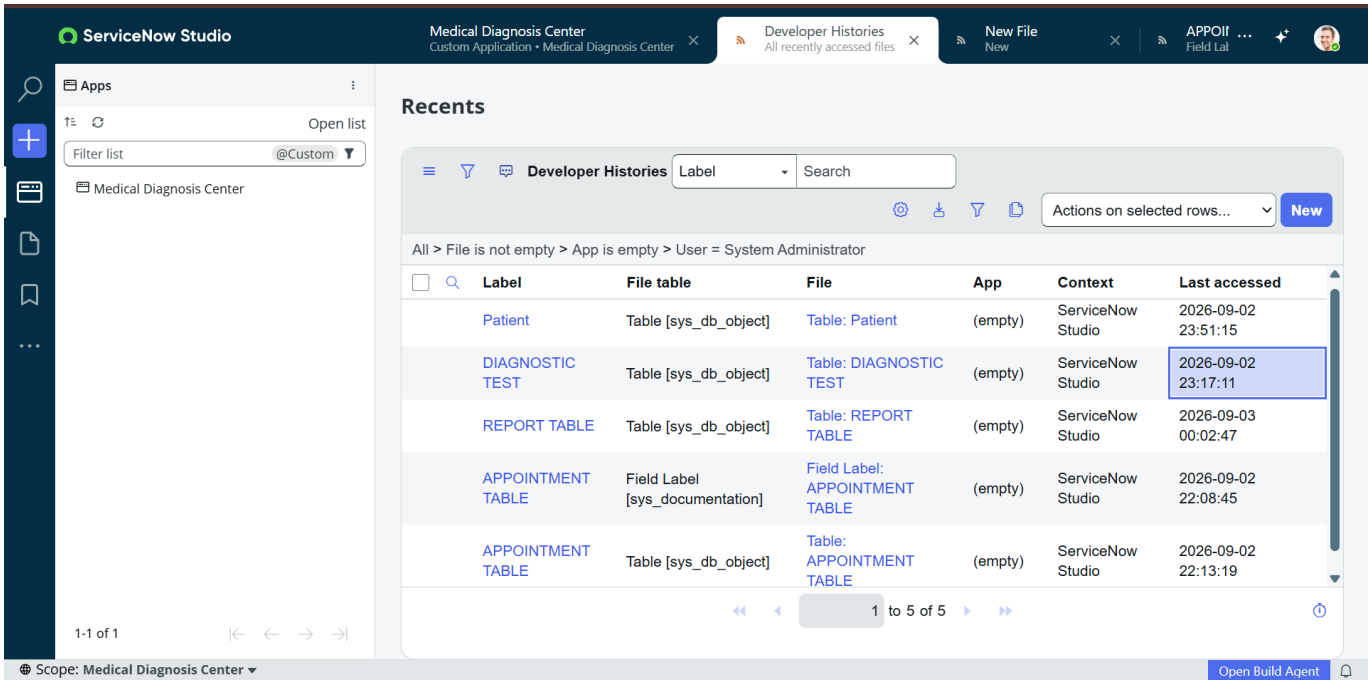
- Activity 1.1: Open the ServiceNow instance
- Sign in to the configured ServiceNow instance and verify that the required roles and application development access are available.
- Activity 1.2: Open the Medical Diagnosis Center application
- Open the application development area and verify the Service Portal, tables, scripts, notifications, ACLs and reporting components required for the project.



Milestone 2: Custom Table Creation and Setup

- Activity 2.1: Create the core Medical Diagnosis Center tables

- Create the Patient table to store patient information.
- Create the Diagnostic Test table to store test details.
- Create the Appointment table to store booking and lifecycle information.
- Create the Report table to store diagnostic report information.
- Create the Payment table to store payment information where the phase-based feature is enabled.



Activity 2.2: Patient and Diagnostic Test Tables

Patient Table

Field / Area	Purpose
Patient Information	Stores required patient details.
Contact Information	Stores contact number and email address.
Patient Identity	Provides the patient record referenced by appointments and reports.

Diagnostic Test Table

Field	Purpose
Test Name	Name of the diagnostic test.
Test Code	Unique configured code for the test.
Category	Diagnostic category.
Description	Description of the test.
Price	Configured test price.
Duration	Configured test duration.
Active Status	Controls whether the test is available.

Activity 2.3: Appointment, Report and Payment Tables

Table	Purpose	Main Relationships
Appointment	Stores booking, date/time, status, approval and payment-related information.	Patient + Diagnostic Test + Payment + Report
Report	Stores diagnostic report information and report lifecycle data.	Patient + Appointment
Payment	Stores payment status and configured transaction information.	Appointment

Data Relationships

Patient → Appointment → Diagnostic Test / Payment / Report

Milestone 3: Service Portal Setup

- Activity 3.1: Create the Service Portal pages

- Home Page
- Test Catalog Page
- Book Appointment Page
- My Appointments Page
- Lab Reports Page
- Payment Page (phase-based)
- Admin Dashboard Page

The Service Portal provides the patient-facing interface for the diagnostic-center workflow.

Diagnostic Test Catalog

The Medical Test Catalog Widget displays available diagnostic tests. Test information can include name, code, category, description, price, duration and active status.

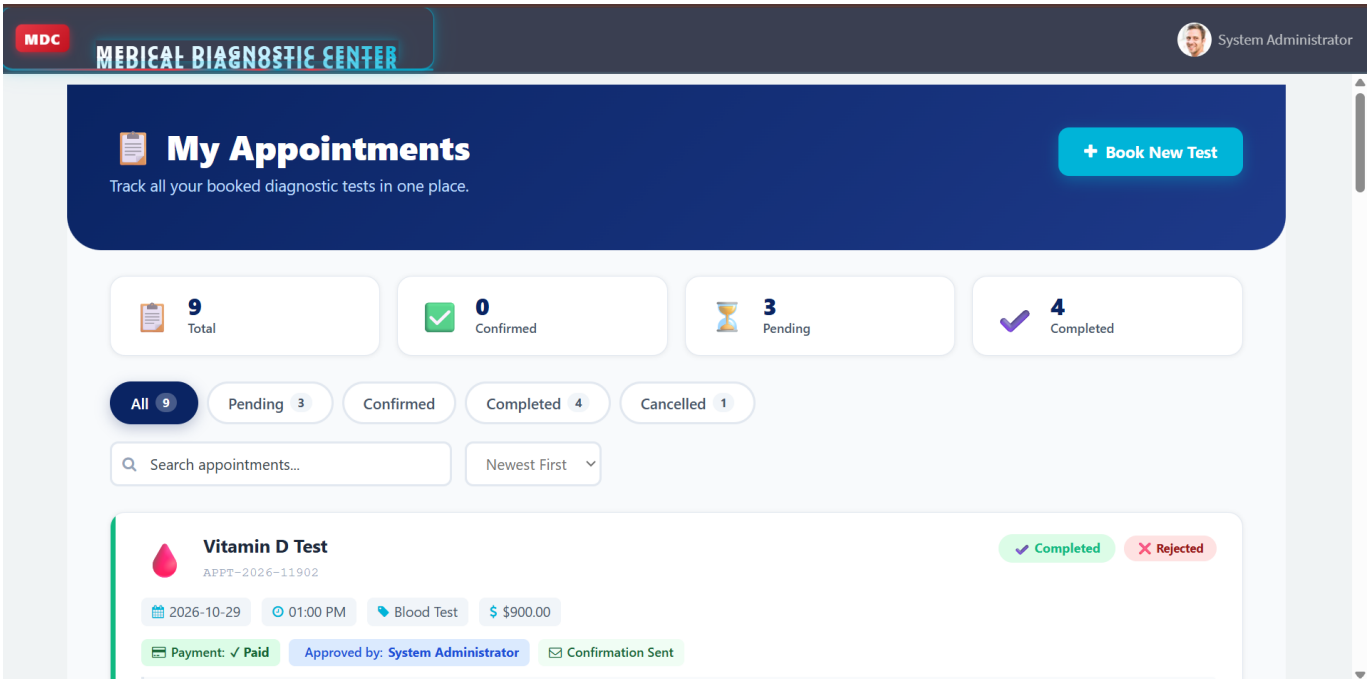
Test Name	Code	Category	Price	Duration
Complete Blood Count (CBC)	CBC-001	Blood Test	₹45	30 minutes
Lipid Profile	LIP-002	Blood Test	₹60	30 minutes
HbA1c (Diabetes)	HBA-003	Blood Test	₹70	30 minutes
Thyroid Function Test (TFT)	THY-004	Blood Test	₹95	30 minutes
Liver Function Test (LFT)	LFT-005	Blood Test	₹75	30 minutes
Kidney Function Test (KFT)	KFT-006	Blood Test	₹65	30 minutes

Appointment Booking

The Appointment Booking Widget allows patients to select a diagnostic test, choose an appointment date and time slot, enter patient details and submit an appointment request.

Patient Details

- Full Name
- Age
- Gender
- Contact Number
- Email Address



Appointment Validation

The system is designed to validate appointment information before a booking is accepted.

- Mandatory-field validation
- Available test slots
- Duplicate booking prevention
- Overlapping appointment prevention
- Equipment conflict prevention
- Technician conflict prevention
- Capacity control per time slot
- Maximum number of tests per patient per day
- Advance booking limits
- Cut-off time restrictions

Example: A time slot can be configured with a maximum number of patients per hour. Once the configured capacity is reached, additional bookings can be prevented.

Milestone 4: Approval and Status Management

Appointments can be managed through configured approval and lifecycle states.

Requested → Pending Approval → Confirmed → Sample Collected → Processing → Completed → Report Available

Alternative lifecycle outcome: Cancelled

Approval handling can record approval notes and Approved By. Rejection handling can record the rejection reason and clear Approved By.

Appointment Status Lifecycle

Status	Purpose
Requested	Initial appointment request.
Pending Approval	Waiting for configured approval.
Confirmed	Appointment approved/confirmed.
Sample Collected	Patient sample collection completed.
Processing	Laboratory processing is in progress.
Completed	Diagnostic processing completed.
Report Available	Report is available to authorized users.
Cancelled	Appointment cancellation outcome.

Milestone 5: Email Notification Setup

Activity 5.1: Approval Notification

Configuration	Value
Name	Approval Notifications
Table	x_1883080_medica_0_appointment
When to Send	Record Inserted or Updated
Condition	Approval Status changes to Approved
Active	True

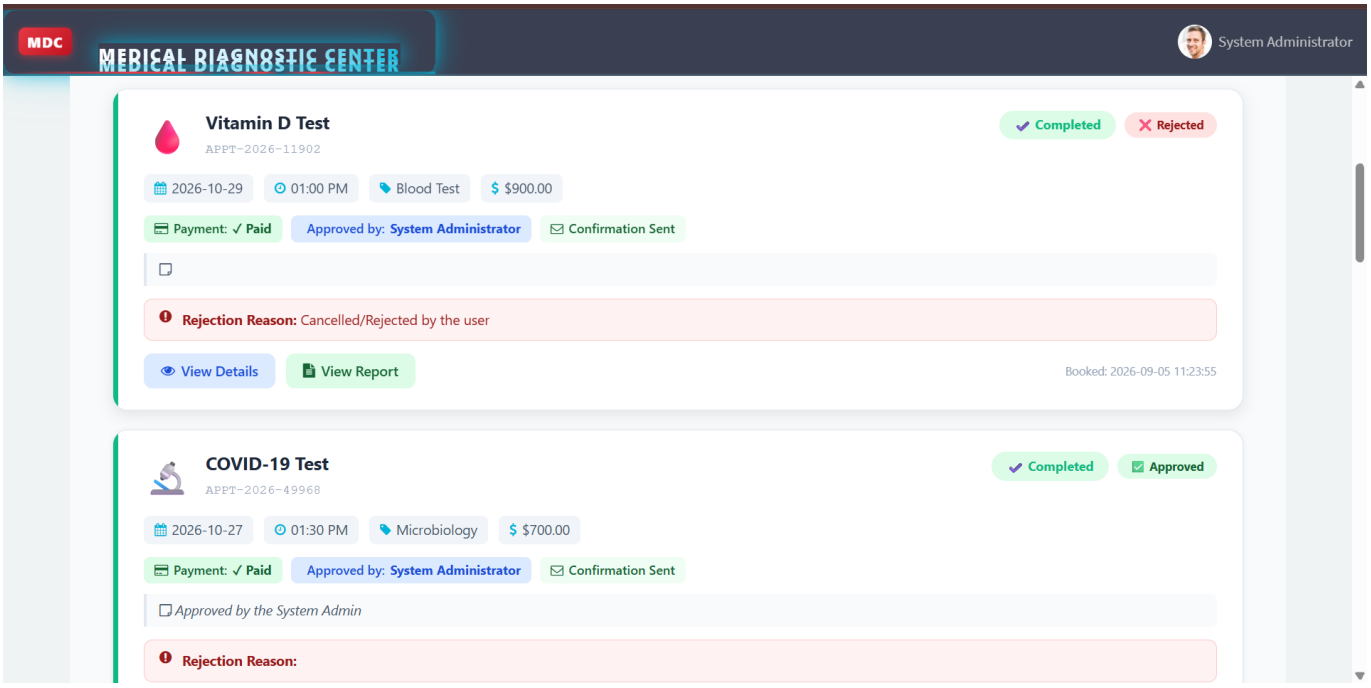
Activity 5.2: Report Notification

Configuration	Value
Name	reportsNotifications
Table	x_1883080_medica_0_report_table
When to Send	Record Inserted

Condition	Report number is Not Empty
Active	True

Diagnostic Report Management

Diagnostic reports are linked to appointments and patients. The report workflow supports report generation after diagnostic processing and makes completed reports available to authorized users.

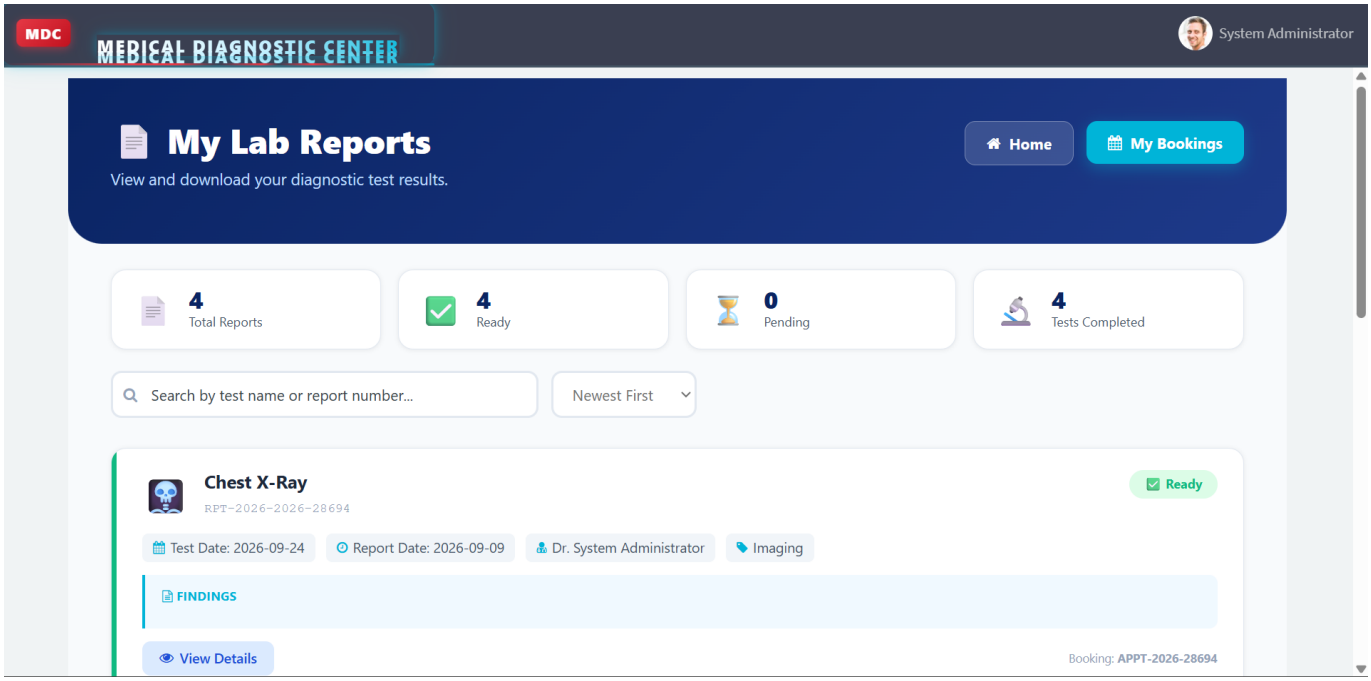
- Create/link report to appointment
- Link report to patient
- Generate report number
- Record report date
- Record verifier
- Set report status
- Expose report through Lab Reports functionality
- 

Report Lifecycle and Lab Reports

Report Lifecycle

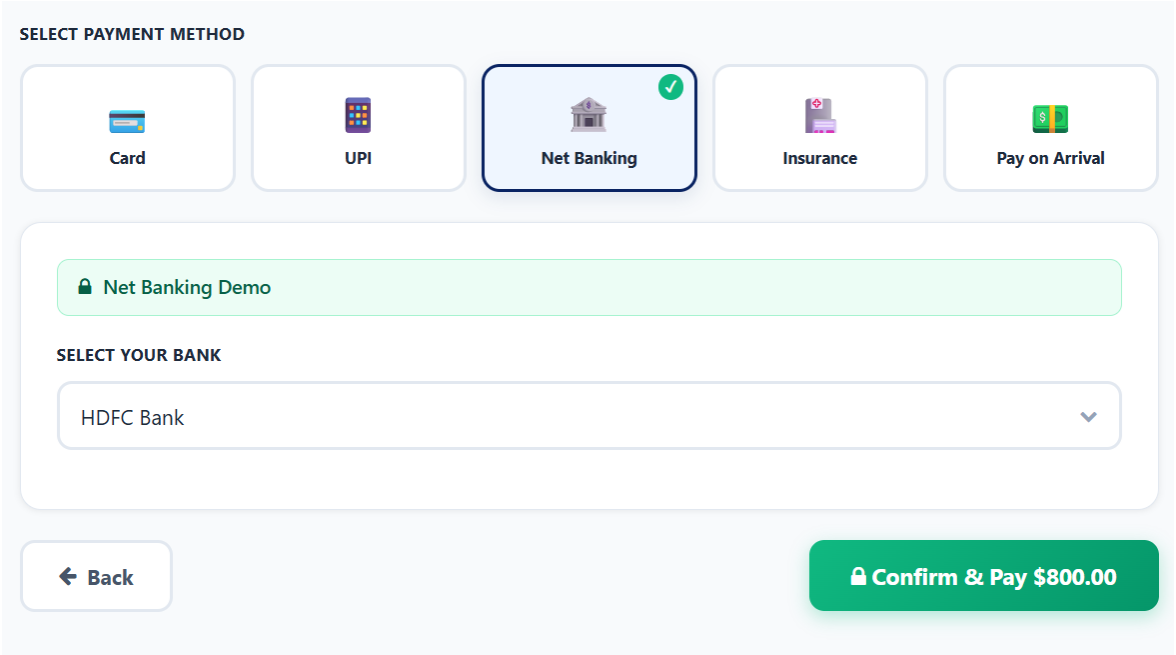
Sample Collected → Processing → Completed → Report Generated → Report Available

The Lab Reports Widget allows patients to view available reports, view report details and download reports where configured.

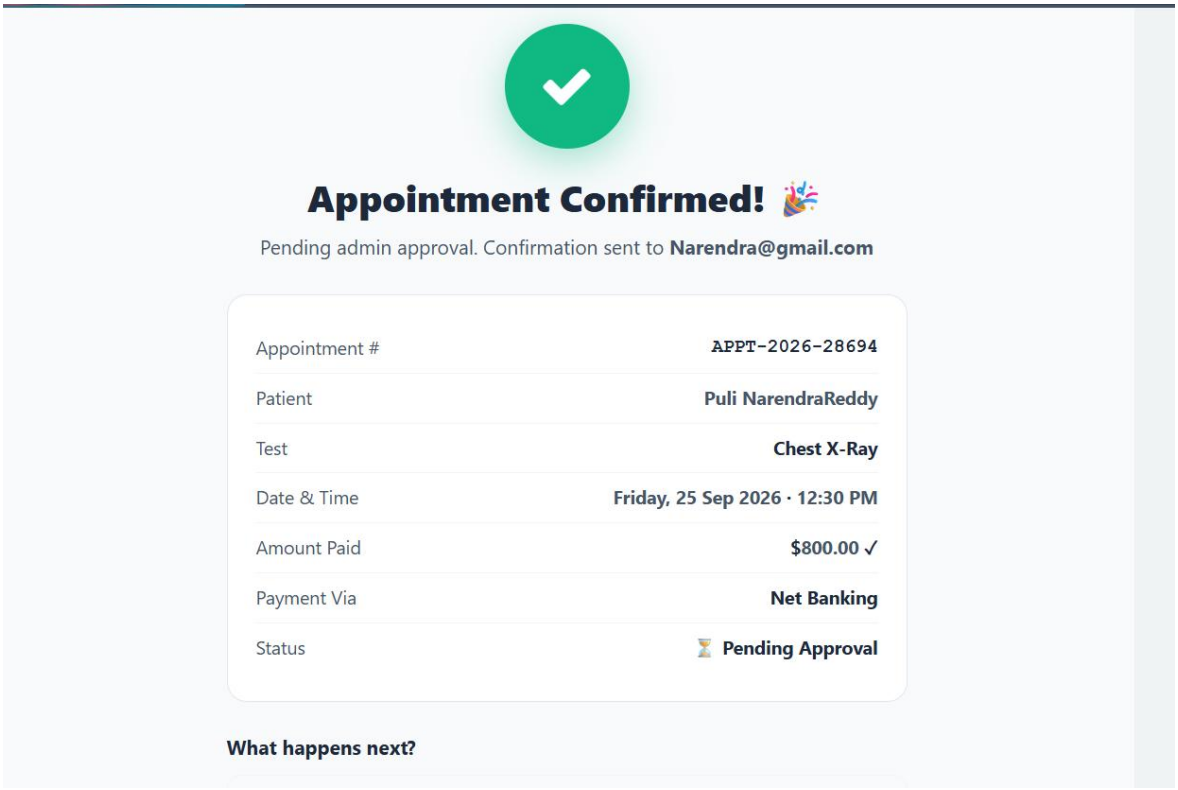


Payment Management

Payment functionality is designed as a phase-based feature. When enabled, it can support test-based pricing, payment status, online payment, digital invoice generation, receipts and transaction history.



Payment Method Selection



- Automatic fee calculation
- Test-type based pricing
- Normal / Express urgency
- Home collection charges
- Online payment
- Payment status tracking
- Digital invoice generation
- Payment receipts
- Transaction history

Payment Status: Paid | Pending | Failed

Cancellation and Refund

The configured cancellation Business Rule applies a 10% deduction from the total amount.

Example

Item	Amount
Total Amount	₹500
10% Deduction	₹50
Refund Amount	₹450

The refund amount is automatically populated when the configured cancellation condition is satisfied.

Business Rules and Automation

1. Auto Populate Refund Amount on Cancellation

When an appointment is cancelled, the Business Rule reads the total amount, calculates the 10% deduction, calculates the remaining refund and stores the refund amount.

2. Auto Create Report on Appointment Complete

When an appointment changes to Completed, the system can create a report if one does not already exist, generate a report number, retrieve diagnostic test information, generate findings, link the report to the appointment and patient, record report date and verifier, and set report status.

3. Rejection Details Updation

When an appointment is rejected, the rejection reason is populated and Approved By is cleared. When approved, the approval note is recorded, Approved By is populated and rejection reason is cleared.

Flow Designer / Workflow Automation

Flow Designer workflows can be used to automate configured diagnostic-center processes such as appointment approval, status transitions, notifications and report-related actions.

- Trigger based on appointment changes
- Validate or route approval
- Update appointment status
- Send configured notifications
- Support report availability communication
- Coordinate lifecycle actions

Milestone 6: Security and ACL Configuration

The application uses Access Control Lists (ACLs) to control access to application data.

- Table-level access control
- Field-level access control
- Role-based access
- Patient-specific record access
- Appointment creation permissions
- Laboratory report access restrictions
- Administrative access restrictions
- Impersonation testing
- Debug Security Rules validation

Roles and Stakeholder Access

Role	Main Responsibilities
Patient	Register, browse tests, book appointments, view status and access authorized reports.
Receptionist	Create/update patient records and coordinate appointments.
Laboratory Technician	Review approved appointments, collect samples and update processing status.
Doctor	Review diagnostic information/report data where configured.
Lab Manager	Review specialized/high-value appointments, approve/reject where configured and monitor status.
System Administrator	Configure tables, forms, portal pages, widgets, Business Rules, ACLs, notifications, reports, dashboards and Update Sets.

Milestone 7: Update Set and Deployment Preparation

The project uses ServiceNow Update Sets for capturing application configuration for deployment.

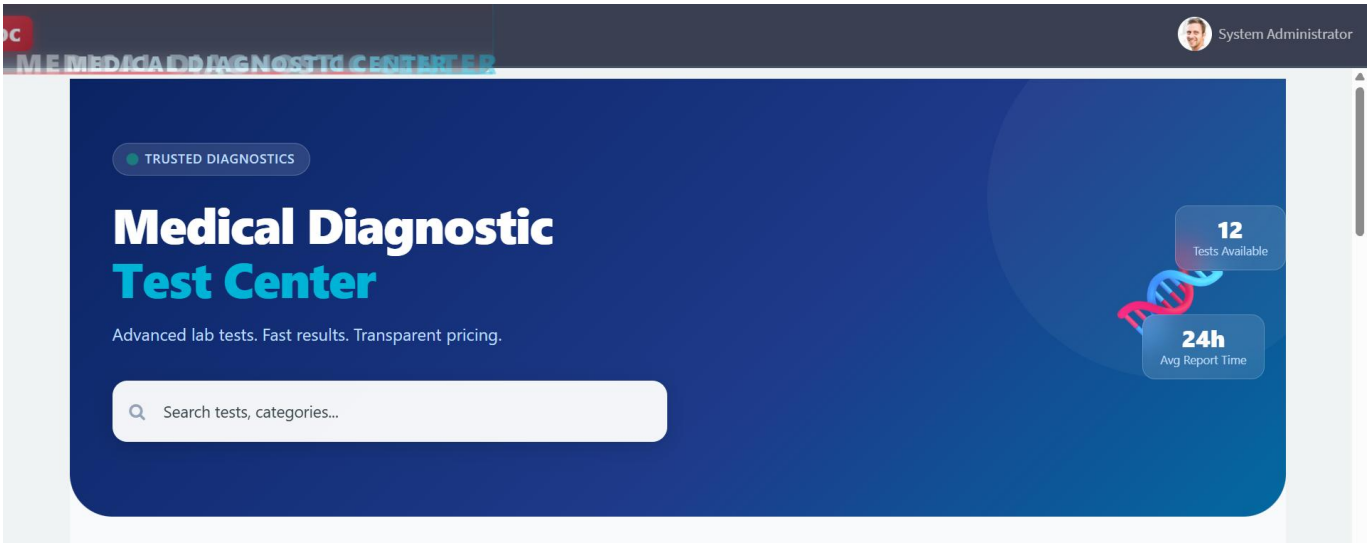
20. Create/update the Update Set
21. Capture relevant application configuration
22. Validate captured records
23. Commit the Update Set
24. Transfer configuration to the target instance
25. Validate the target instance
26. Perform smoke testing
27. Monitor system logs and application behavior

Application Development Components

Technology / Component	Usage
ServiceNow	Application platform
Service Portal	Patient-facing interface
JavaScript	ServiceNow scripting
GlideRecord	Server-side record operations
HTML	Portal UI
CSS	Portal styling
AngularJS	Service Portal client-side development
Business Rules	Automation
Flow Designer	Workflow automation
ACLs	Security
Reports & Dashboards	Analytics
Update Sets	Deployment
Git / GitHub	Version control

Service Portal Pages

Page	Purpose
Home Page	Main landing page for the Diagnostic Center
Test Catalog Page	Displays available diagnostic tests
Book Appointment Page	Allows patients to book diagnostic tests
My Appointments Page	Displays patient appointment details and status
Lab Reports Page	Allows patients to view/download diagnostic reports
Payment Page	Handles payment-related functionality as a phase-based feature
Admin Dashboard Page	Provides administrative reports and analytics



ServiceNow Widgets

1. Medical Test Catalog Widget

Displays configured diagnostic tests and their details.

2. Appointment Booking Widget

Supports test selection, appointment date/time selection, patient details and booking validation.

3. My Appointments Widget

Displays appointments, status, approval status and permitted cancellation/rescheduling actions.

4. Lab Reports Widget

Displays available diagnostic reports and supports viewing/downloading where configured.

Medical Test Catalog Widget

Test Information

- Diagnostic Test Name
- Test Code
- Test Category
- Description
- Price
- Duration
- Active Status

Configured examples include CBC, Lipid Profile, HbA1c, Thyroid Function Test, Liver Function Test, Kidney Function Test, Urine Routine & Microscopy, Chest X-Ray, ECG, COVID-19 PCR Test, Vitamin D Test and Iron Studies.

Appointment Booking Widget

The Appointment Booking Widget allows patients to:

- Select a diagnostic test
- Select an appointment date
- Select a time slot
- Enter patient details
- Submit an appointment request
- Validate mandatory information
- Check booking availability

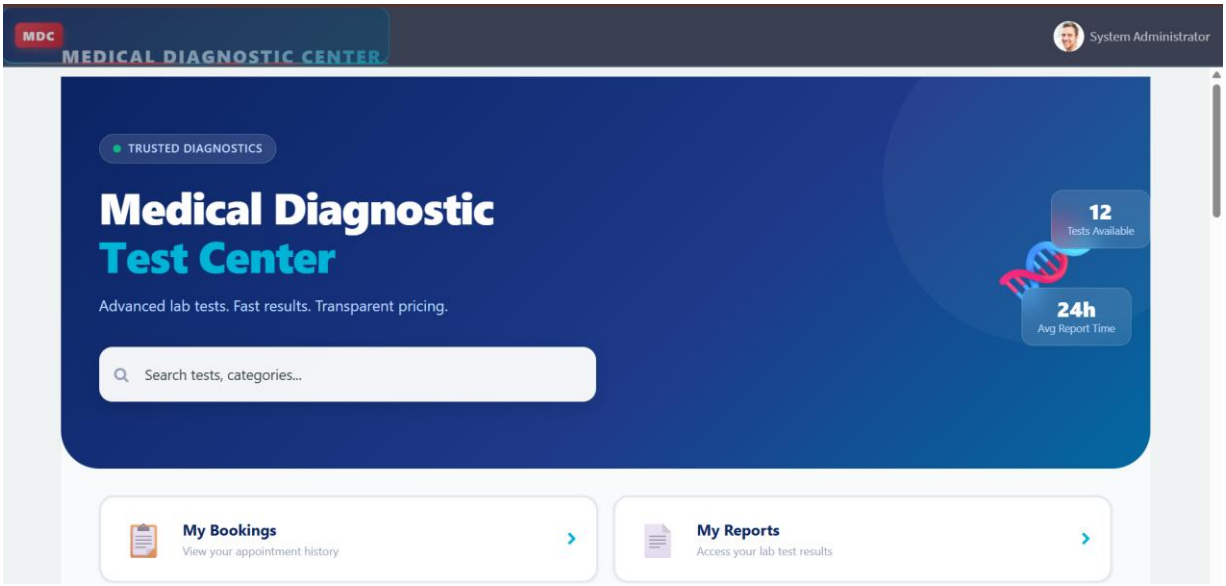
Milestone 8: Testing and Deployment Validation

Activity 8.1: Conduct functional testing to verify patient registration, diagnostic test selection, appointment booking, validation, approval, laboratory processing, report generation, notifications, cancellation/refund, ACL security and dashboard reporting.

Test ID	Scenario	Expected Result	Status
TC-001	Register patient	Patient record created	Pending / to be recorded
TC-002	Book diagnostic test	Valid appointment saved; conflicts blocked	Pending / to be recorded
TC-003	Approve appointment	Confirmed status and notification	Pending / to be recorded
TC-004	Process appointment	Lifecycle progresses correctly	Pending / to be recorded
TC-005	Generate report	Report linked to appointment/patient	Pending / to be recorded
TC-006	Cancellation/refund	10% deduction applied	Pending / to be recorded
TC-007	ACL access	Unauthorized access denied	Pending / to be recorded

Home Page:

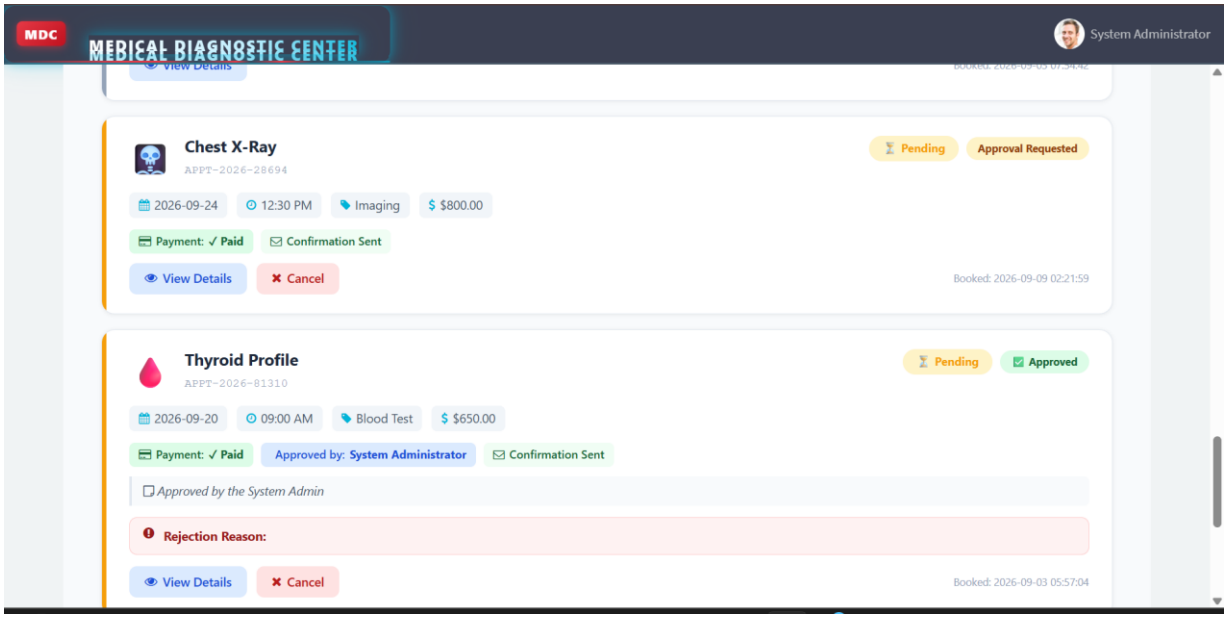
Service Portal Home Page



The Home Page acts as the main landing page for the Medical Diagnosis Center Service Portal.

Dashboard Page:

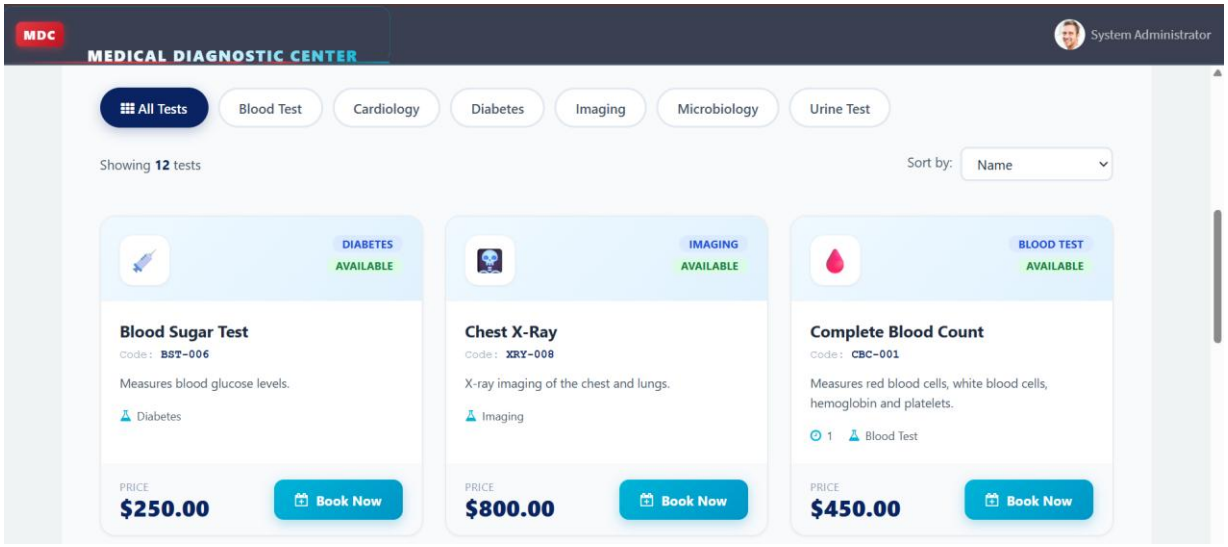
My Bookings



The patient dashboard can display appointment information, status and configured actions.

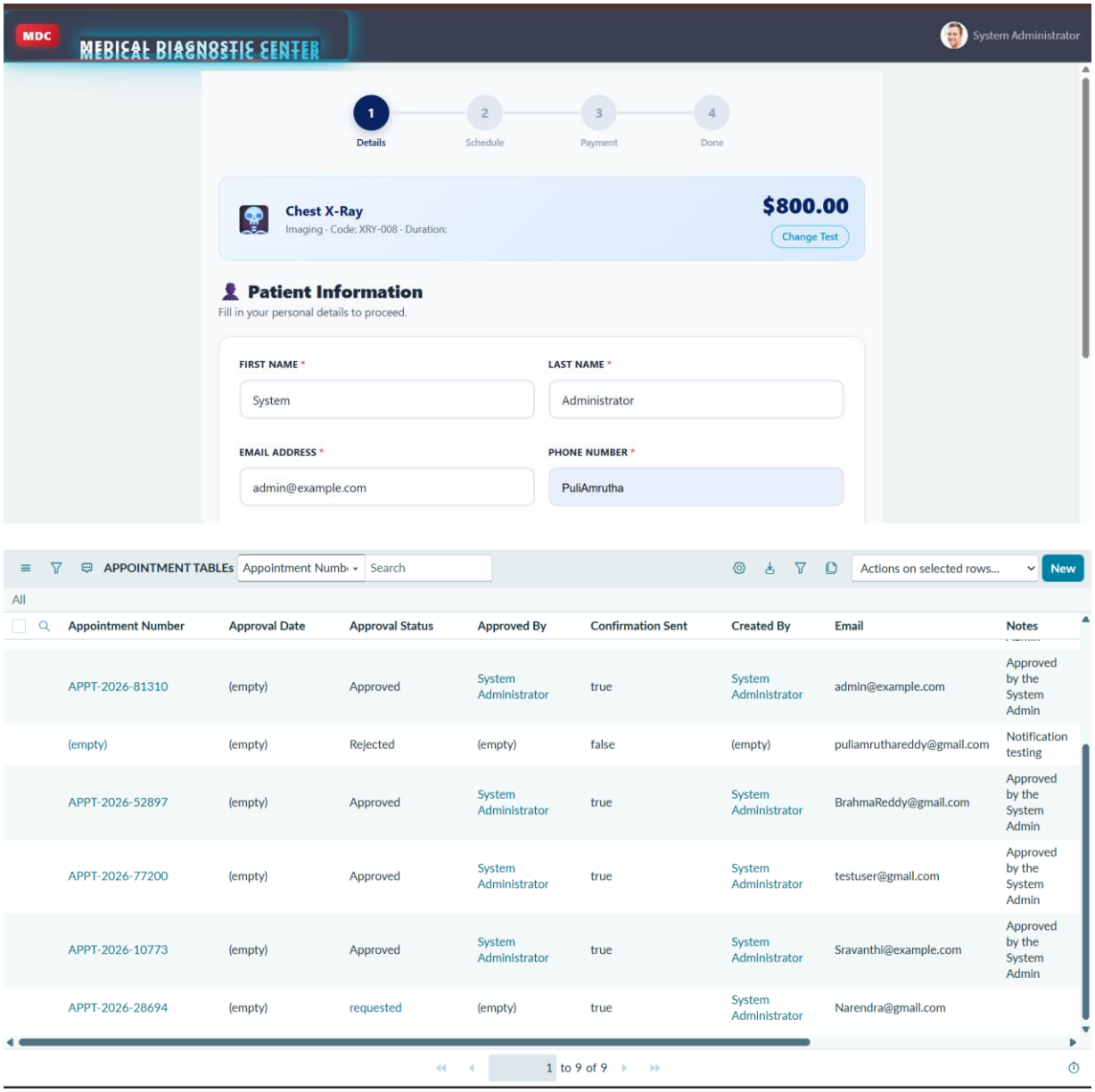
Test Catalog Page:

Medical Test Catalog



Book Appointment Page:

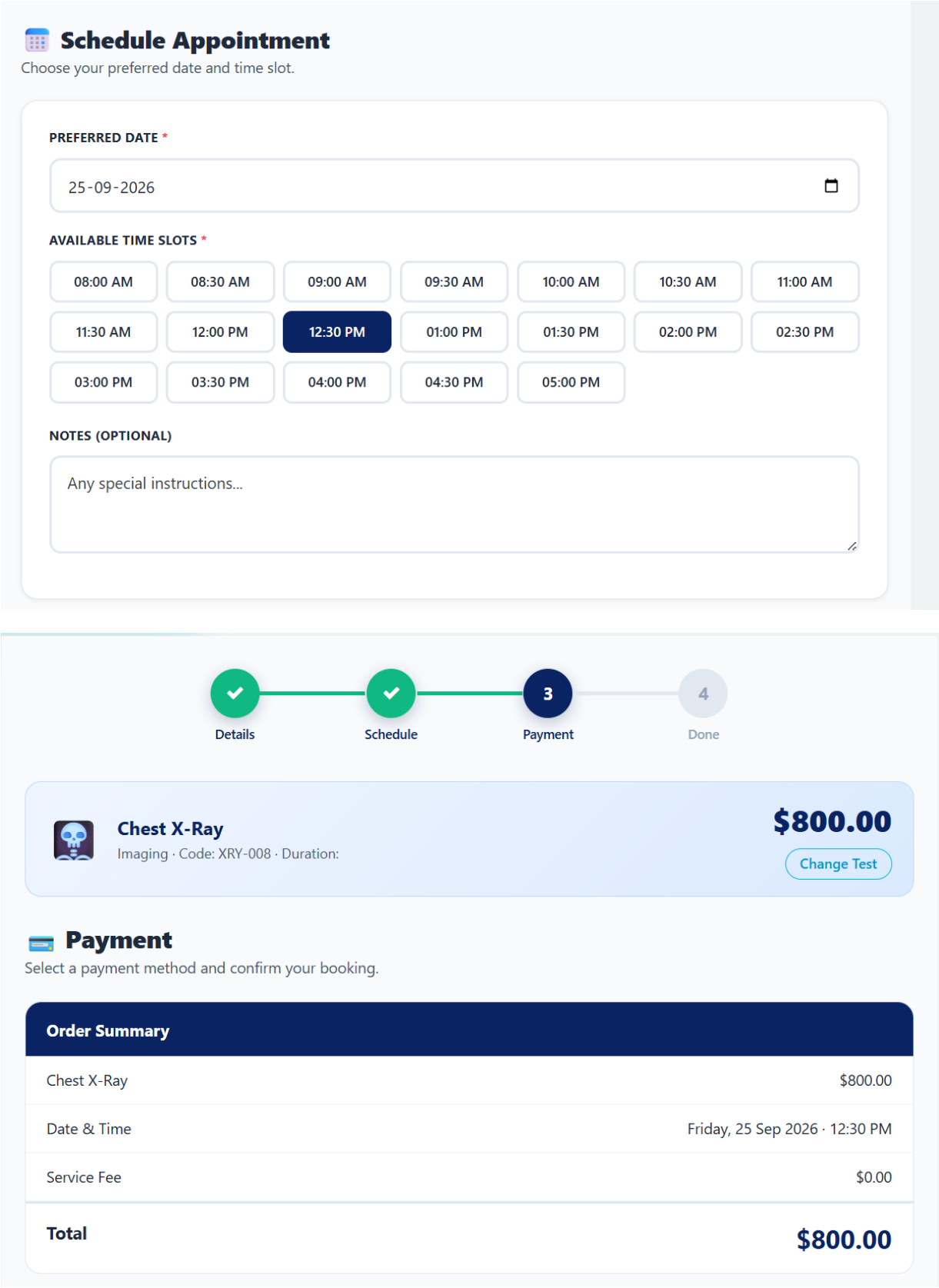
Book Appointment



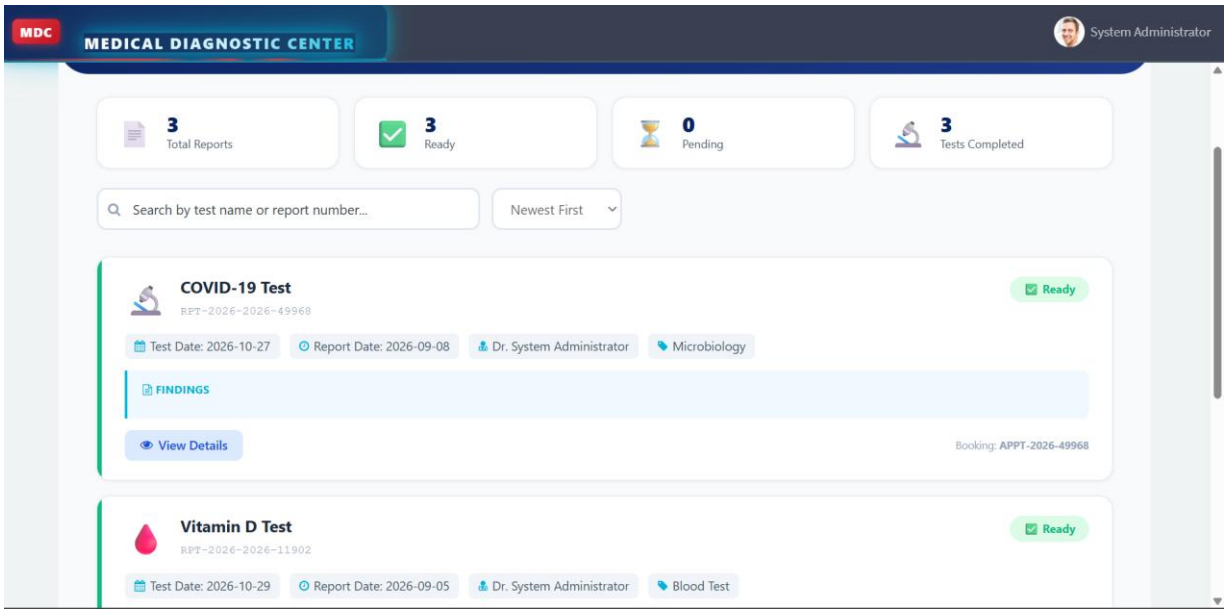
Insert the implemented Service Portal screenshot showing diagnostic test selection, date/time selection and booking controls.

Appointment Details / Validation Page:

Appointment Details

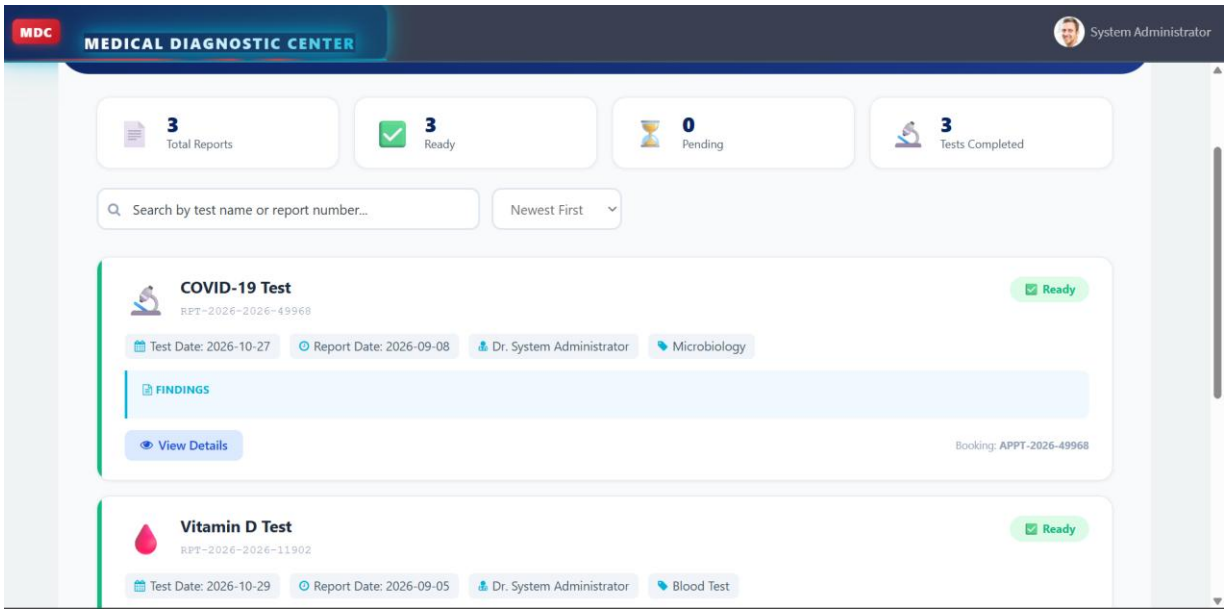


Approval and Processing Page:



Lab Reports Page:

My Reports



Insert the implemented report view showing the authorized patient's diagnostic report.

Payment / Cancellation Page:

payment details

If payment is enabled in the implemented phase, insert the payment evidence. Otherwise insert the configured cancellation/refund record.

Health Reports Dashboard:

My Reports

Dashboard Name: Health Reports Dashboard

Appointments Reports can group appointment data by Status and Approval Status, with configured aggregation and dashboard placement.

ServiceNow Data and Notifications

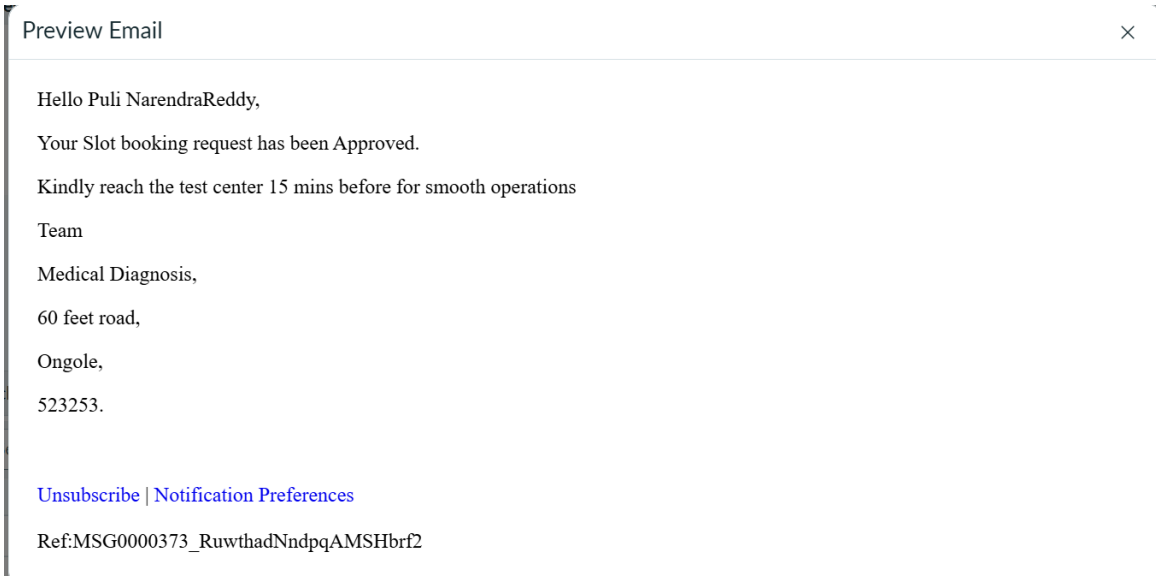
1. Appointment Records

Appointment records contain patient, diagnostic test, date/time, status, approval and configured payment/report information.

2. Diagnostic Report Records

Report records are linked to the relevant appointment and patient and can become available after completion.

3. Mail to the User



Email Notification Preview

Configured email notifications can communicate appointment approval and report availability. Additional scenarios include booking confirmation, payment confirmation, rejection, reminders, cancellation and refund notifications where configured.

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Project: Medical Diagnosis Center in ServiceNow

Conclusion:

The Medical Diagnosis Center has been designed as a centralized ServiceNow workflow for patient registration, diagnostic test selection, appointment management, laboratory processing, report delivery, notifications, security and operational reporting. The solution connects these activities through custom tables, Service Portal pages and widgets, Business Rules, Flow Designer workflows, ACLs, notifications and dashboards.

The configured appointment lifecycle supports Requested, Pending Approval, Confirmed, Sample Collected, Processing, Completed and Report Available states, with Cancellation as an alternative outcome. The project documentation also defines validation rules, report management, phase-based payment/refund handling, role-based security and management reporting.

The solution is intended to reduce manual paperwork and duplicate data entry, centralize patient and diagnostic information, improve appointment coordination, automate configured approval and notification processes, protect sensitive records through access controls and provide operational visibility through reports and dashboards.

Final execution screenshots and actual test results should be inserted from the implemented ServiceNow instance. The supplied project materials explicitly leave execution measurements pending where they were not provided.

Project Outcome

The Medical Diagnosis Center provides a structured ServiceNow workflow for managing diagnostic-center operations from patient registration through appointment booking, approval, laboratory processing and report availability.